

**Project Interim Report**  
**On**  
**Expense Tracker System**  
**(Using Core Java & OOP Concepts)**

**RU-100-01-00012**  
**Object Oriented Programming with Java**

**SESSION: 2025-26**



Submitted By  
**Priyanshu Yadu**  
**RU-25-11054**

**Bachelor of Technology in Computer Science  
& Engineering in association with Google**

Under the Guidance of  
**Mr. Ashish Shivhare**

**SCHOOL OF COMPUTER SCIENCE AND ENGINEERING**  
**RUNGTA INTERNATIONAL SKILLS UNIVERSITY**

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## ➤ Certificate

This is to certify that the project titled “**Expense Tracker System Using OOP**” has been successfully completed by **Priyanshu Yadu** under the guidance of faculty.

(Signature of HOD)

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## ➤ Acknowledgement

I would like to express my sincere gratitude to my faculty members and my college for providing me with the opportunity to work on this project. Their guidance and support helped me in understanding Object-Oriented Programming concepts practically. I also thank my friends and mentors for their valuable suggestions during the development of this project.

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## ➤ ABSTRACT

The Parking Lot Management System is a software-based solution designed to efficiently manage parking spaces for different types of vehicles such as bikes, cars, and trucks. In real-world scenarios, manual parking management often leads to confusion, inefficient space utilization, and difficulty in tracking vehicle entry and exit. This project addresses these issues by automating the parking process using object-oriented programming concepts.

The system assigns available parking slots to vehicles based on their type and tracks their entry and exit time. It also maintains records of occupied and free slots, ensuring better organization and utilization of space. Additionally, the system can calculate parking fees based on the duration of parking, making it more practical and realistic.

This project is useful for understanding real-world system design and implementing core programming concepts such as classes, objects, arrays, and methods. It provides a simple yet effective simulation of a parking management system that can be further enhanced with advanced features like database integration, GUI, and multi-level parking.

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## ➤ INTRODUCTION

The Parking Lot Management System is developed to simplify and automate the process of managing parking spaces. In crowded areas such as malls, offices, and public parking areas, manual systems are inefficient and prone to errors. This project aims to provide a structured and automated solution.

The system allows users to park and remove vehicles while maintaining accurate records of parking slots. It categorizes vehicles into three types: Bike, Car, and Truck, and assigns slots accordingly. The system ensures that each vehicle is parked in a suitable slot and prevents duplication of vehicle entries.

Key Features:

- Automatic slot allocation based on vehicle type
- Tracking of available and occupied slots
- Vehicle entry and exit management
- Optional parking fee calculation
- Simple and user-friendly interface

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## ➤ OBJECTIVES

- Manage parking slots efficiently
- Track parked and removed vehicles
- Reduce manual work and human errors
- Implement real-world logic using programming
- Apply Object-Oriented Programming (OOP) concepts

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## ➤ SCOPE

This project is suitable for small-scale parking areas such as colleges, offices, or small malls. It can be extended in the future by integrating a database for storing records permanently, adding a graphical user interface, and implementing multi-level parking systems.

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## ➤ SYSTEM REQUIREMENTS

### Hardware:

- Computer/Laptop with minimum 4GB RAM

### Software:

- Java JDK
- IDE (VS Code / Eclipse / Edit Plus)

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## ➤ METHODOLOGY

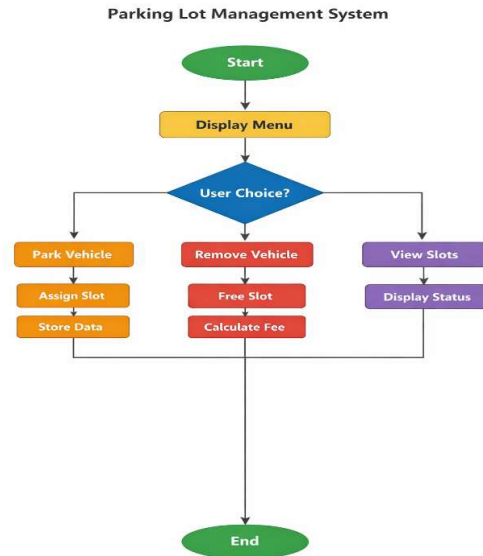
The system works in a step-by-step process:

1. User selects an option (Park / Remove / View Slots)
2. If parking:
  - Enter vehicle ID and type
  - System searches for available slot
  - Assigns slot and stores entry time
3. If removing:
  - Enter vehicle ID
  - System finds vehicle
  - Removes it and frees slot
4. Display available and occupied slots

5. (Optional) Calculate parking fee based on time

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## ➤ FLOWCHART



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## ➤ CLASS DIAGRAM

**Class: Vehicle**

**Variables:**

- vehicleId
- vehicleType
- entryTime

**Methods:**

- Constructor
- Getters

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**Class: ParkingLot**

**Variables:**

- slots[] (array of vehicles)
- slotType[]

**Methods:**

- parkVehicle()
- removeVehicle()

- displaySlots()

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## ➤ IMPLEMENTATION

- **Vehicle Class:**  
Stores vehicle details such as ID, type, and entry time.
- **ParkingLot Class:**  
Manages all parking operations using arrays.
- **parkVehicle():**  
Finds a suitable empty slot and assigns the vehicle.
- **removeVehicle():**  
Searches vehicle by ID, removes it, and frees the slot.
- **displaySlots():**  
Shows total, occupied, and available slots.

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## ➤ SAMPLE INPUT / OUTPUT

1. Park Vehicle
2. Remove Vehicle
3. View Slots

Enter Choice: 1  
Enter Vehicle ID: Bike-101  
Enter Type: Bike  
Vehicle parked in Slot 1

Enter Choice: 2  
Enter Vehicle ID: Bike-101  
Vehicle removed  
Parking Fee: ₹50

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## ➤ FUTURE ENHANCEMENTS

- GUI using Java Swing
  - Database integration (MySQL)
  - Online booking system
  - Multi-level parking
  - Mobile app integration
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## ➤ CONCLUSION

The Parking Lot Management System successfully demonstrates the use of Java and Object-Oriented Programming concepts. It provides an efficient way to manage parking slots and reduces manual effort. The system is simple, functional, and can be expanded into a real-world application with additional features.

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## ➤ REFERENCES

- Java Documentation
- Online Tutorials
- OOP Textbooks



